Bioinformatics

2025/2026

Day 1

Note: /opt/anaconda3/bin/conda init

Exercise 1: Create a dedicated conda environment named bioinfo_lab1 to ensure reproducibility.

- **Requirements:** The environment must include python (version 3.9+), entrez-direct (for data fetching), emboss (for validation), make, and jupyterlab.
- **Deliverable:** Export the environment configuration to a YAML file (environment.yml).

Exercise 2: Download the full genome of *Enterobacteria phage lambda* (Accession: NC_001416).

- **Preferred Method:** Use NCBI's **EDirect** (efetch) to request the data directly from the nuccore database in FASTA format.
- **Alternative Method:** If EDirect fails, use wget or curl to fetch from the NCBI FTP site.
- **Output:** Save the file as lambda_phage.fasta.

Exercise 3: Develop a Python tool to calculate the Reverse Complement of a DNA sequence.

3.1 Create a notebook named prototyping.ipynb. Implement a function reverse_complement(seq) that takes a DNA string and returns its reverse complement.

- **Constraint 1:** The function must handle **case insensitivity** (e.g., treat 'a' and 'A' identically) and return the result in UPPERCASE.
- **Constraint 2:** Handle Ambiguity Codes. Ensure N complements to N.
- **Constraint 3:** Implement a test_reverse_complement() function that validates your logic against at least 3 manual test cases (e.g., ATGC $\rightarrow$ GCAT).

3.2 FASTA Parsing **-** In the same notebook, write a function read_fasta(filepath).

- The function must parse a standard FASTA file, separating the **header** (line starting with >) from the **sequence**.

- **Crucial:** FASTA files often split long sequences across multiple lines. Your parser must concatenate these lines into a single string before processing.

3.3 Convert your validated notebook logic into a command-line script named revcomp.py.

- **Usage:** The script must accept the input file as a command-line argument: Bash: `python revcomp.py lambda_phage.fasta`
- **Output:** Print the header and the transformed sequence to stdout in valid FASTA format.

Exercise 4: Generate a reference dataset using the revseq tool from the EMBOSS suite.

- **Command:** Use revseq on lambda_phage.fasta.
- **Tip:** Use the -notag flag to prevent EMBOSS from appending metadata to the header, simplifying comparisons in the next step.

Exercise 5: Compare the output of your revcomp.py against the EMBOSS revseq output.

- **Challenge:** A direct diff may fail even if the sequences are identical, because EMBOSS wraps sequence lines at 60 characters by default, whereas your script may print a single long line.
- **Tip:** Construct a shell command (using grep, tr, or awk) to strip headers and newlines from both outputs, then compare the **MD5 checksums** of the raw sequence strings to prove equivalence.

Exercise 6: Wrap the entire pipeline in a Makefile (Orchestration).

- **Targets:** The Makefile should include rules to:
    1. Download the data (lambda_phage.fasta).
    2. Run revcomp.py on the downloaded data.
    3. Run the EMBOSS validation.
    4. Compare the results and print "Validation Passed" or "Validation Failed."